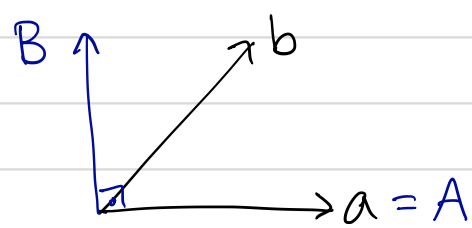
Lecture 26

- QR decomposition
- Eigenvalue computations
- Fourier series

Gram-Schmidt:

$$a,\ b,\ c \longrightarrow \underset{q_1}{\underset{\parallel}{\frac{A}{\|A\|}}},\ \underset{q_2}{\underset{\parallel}{\frac{B}{\|B\|}}},\ \underset{q_3}{\underset{\parallel}{\frac{C}{\|C\|}}}$$

independent $\qquad$ orthonormal



$$A = a$$

$$B = b - \frac{A^T b}{A^T A} A$$

$$C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B}$$

$$\begin{bmatrix} | & | & | \\ a & b & c \\ | & | & | \end{bmatrix} = \begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ \cancel{q_2^T a}^{\,0} & q_2^T b & q_2^T c \\ \cancel{q_3^T a} & \cancel{q_3^T b}^{\,0} & q_3^T c \end{bmatrix}$$

$$A = Q \cdot R$$

$$QQ^T = I \Rightarrow A = Q\underbrace{Q^T A}_{R}$$

A matrix with

$$|\lambda_1| > |\lambda_2| > \cdots > |\lambda_n|$$

Power method (computes λ_1)

Start: v_0 (length 1)

Step: $v_{i+1} = A v_i / \|A v_i\|$

For most v_0's:

$$v_{i+1}^T A v_{i+1} \xrightarrow{i \to \infty} \lambda_1$$

Idea: $x_1, \ldots, x_n$ eigenvectors

$$v_0 = c_1 x_1 + \cdots + c_n x_n$$

$$A^k v_0 = c_1 \lambda_1^k x_1 + \cdots + c_n \lambda_n^k x_n$$

$$= c_1 \lambda_1^k \left(x_1 + \cancel{\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k x_2 + \cdots}^{\,0} \right)$$

QR method (computes all λ's)

Start: Q_0 (orthogonal)

Step: $A Q_i = Q_{i+1} R_{i+1}$

For most Q_0's:

$$\underbrace{Q_i^T A Q_i}_{T_i} \xrightarrow{i \to \infty} T \text{ triangular}$$

λ's on diagonal

QR method (actually)

Computes T_{i+1} from T_i

Start: $\tilde{Q}_0$ orthogonal, $T_0 = \tilde{Q}_0^T A \tilde{Q}_0$

Step: $T_i = \tilde{Q}_{i+1} R_{i+1}$

$T_{i+1} = R_{i+1} \tilde{Q}_{i+1}$

$(\tilde{Q}_i = Q_{i-1}^T Q_i)$

For most Q_0:

$T_i \longrightarrow T$ triangular

λ's on diagonal

Projections with orthonormal basis

$q_1, q_2, \dots, q_n$

Any

$$v = x_1 q_1 + x_2 q_2 + \dots + x_n q_n$$

$$q_1^T v = x_1 \underbrace{q_1^T q_1}_{1} + 0 + \dots + 0$$

$$v = Qx \implies x = Q^{-1} v = Q^T v$$

Fourier series

Periodic function: $f(x) = f(x+2\pi)$

$$f(x) = a_0 \cdot \underline{\underline{1}} + a_1 \underline{\underline{\cos x}} + b_1 \underline{\underline{\sin x}} + a_2 \underline{\underline{\cos 2x}} + b_2 \underline{\underline{\sin 2x}} + \dots$$

Vectors:

$$v^T w = v_1 w_1 + v_2 w_2 + \dots + v_n w_n$$

Functions:

$$f^T g = \int_0^{2\pi} f(x) g(x)\, dx$$

$\sin x \perp \cos x$:

$$\int_0^{2\pi} \sin x \cos x \, dx = \frac{1}{2} (\sin x)^2 \Big|_0^{2\pi} = 0$$

Find: a_1

$$\int_0^{2\pi} f(x)\, dx = a_1 \overbrace{\int_0^{2\pi} \cos(x)^2 dx}^{\pi} + \underline{\underline{0}}$$